

Optical Measurements of Plasma Dynamics in Carbon Fiber Z-Pinches

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Abstract—A series of experiments has been carried out on the Mega Ampere Generator for Plasma Implosion Experiments (MAGPIE) generator in order to study the dynamics of carbon fiber Z-pinches. The generator was operated at 1.4 MV, with a peak current of 1 MA, and a rise time of 150 ns. In some shots, a current prepulse of about 30 kA was provided to study its influence on the dynamics of the fiber pinch. Carbon fibers of 7, 33, and 300 μm diameter were used during these experiments. The diagnostics employed were a self-referencing interferometer, a two-frame Schlieren system, an optical streak camera, and a four-frame X-ray framing camera. A novel feature of these measurements is the employment of an optical streak camera with a set of four slits arranged along the fiber axis and displaced in the radial direction. This permitted the study of the temporal evolution (axial and radial) of the plasma regions emitting in the visible part of the spectra. Correlation between these regions of the plasma and the location of X-ray hot spots is discussed. In carbon fibers of 33 μm diameter, the radial expansion velocity measured from Schlieren images was $3.6 \times 10^6 \text{ cm/s}$ and $5.5 \times 10^6 \text{ cm/s}$ for shots with and without prepulse, respectively. The dominant axial wavelengths of instabilities in the coronal plasma were between 0.05 and 0.2 cm, which correspond to ka values between 10 and 20, where k is the wavenumber of the instability and a is its amplitude. The dynamics of carbon fibers of different diameters are compared.

Index Terms—Carbon, plasma measurements, plasma pinch, plasmas, pulse power system.

I. INTRODUCTION

DENSE Z-pinches formed from cryogenic fibers of deuterium and tritium have been suggested as a way to achieve nuclear fusion [1], [2] and radiative collapse [3]–[6]. To reach radiative collapse the pinch plasma must be confined for about 100 ns and the pinch current must exceed the Pease–Braginskii limit [7], [8] which is about 1 MA for hydrogen. The stability properties of the dense Z-pinch plasma at high current are thus of great importance and still have not been fully understood. Results of computer simulations [9], [10] and experiments [11], [12] suggest that the fiber Z-pinch consists of two different regions: a low density hot corona plasma and a dense solid core which remains for a long time during the discharge. The $m = 0$ MHD instability starts to grow in the corona plasma, with a growth time which is short due to the low density and high temperature of this plasma [13], [14]. Interaction of this hot corona plasma with

the solid core of the fiber determines the subsequent dynamics of the pinch. The presence of a dense solid core is essential for interpretation of experimental observations.

Pre-ionization of an outer layer of the fiber changes the parameters of the coronal plasma and thus affect the dynamics of the pinch. The plasma generated by the prepulse surrounding the fiber creates a density profile in which the Kadomtsev criteria is satisfied at most radial positions. Two-dimensional simulations show a delay in the onset of instabilities but at later time the plasma is dominated by the $m = 0$ instability [15]. Influence of the current prepulse on the development of instabilities in deuterated polyethylene (CD₂) fibers has been reported [16]. Experiments carried out using carbon fibers in which the time between the current prepulse (7 kA) and pinch current (up to 150 kA) was varied showed a delay in the development of instabilities [17].

The present experiments studied the temporal evolution of the corona plasma in a carbon fiber Z-pinch at currents up to 1 MA. The pinch dynamics were compared for fibers of different diameters and the influence of the current prepulse on the parameters of the corona plasma was observed.

II. EXPERIMENTAL SETUP AND DIAGNOSTICS

The experiments were carried out on the Mega Ampere Generator for Plasma Implosion Experiments (MAGPIE) generator [18], operated at 1.4 MV, 1 MA, and 150 ns rise time. The generator is composed of 4 pulse forming lines (PFL), which are coupled to a final transfer line through gas filled trigatron switches. In order to investigate the influence of a small current prepulse (30 kA) applied before the main discharge, a resistor (75 or 170 ω) was located such that it bypassed the trigatron switch in one of the PFL's of the generator.

The machine was fired with a base pressure in the discharge chamber of less than 10^{-4} mbar, which ensured a reliable performance of the magnetically insulated transmission line (MITL). A schematic of the discharge chamber and the relative location of the different diagnostics are shown in Fig. 1. The current through the plasma was monitored with two Rogowski coils. The first coil was located in one of the eight current return posts while the second was placed in the main return current path. The voltage was measured with a capacitive probe located in the upper part of the transfer line. A neodymium-yttrium aluminum garnet (Nd-YAG) Q-switched laser, frequency-doubled to 532 nm and SBS pulse-compressed to 400 ps [19], was employed in the optical probing. The beam was separated into two perpendicularly polarized beams and one of them was delayed by 3 ns, with respect to the other. Both beams followed identical

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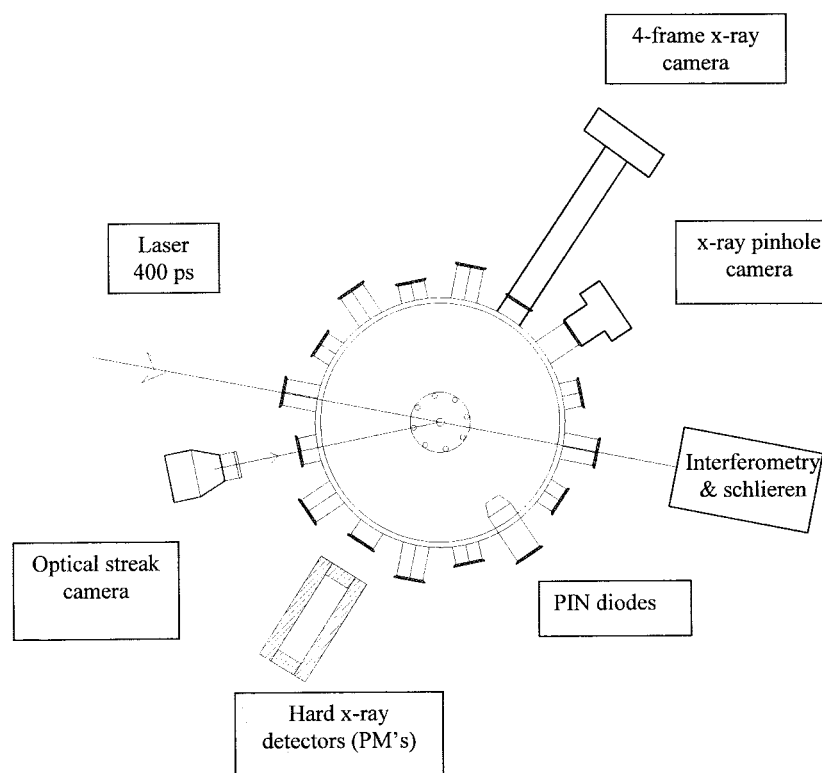


Fig. 1. Schematic of the diagnostics in the discharge chamber of MAGPIE.

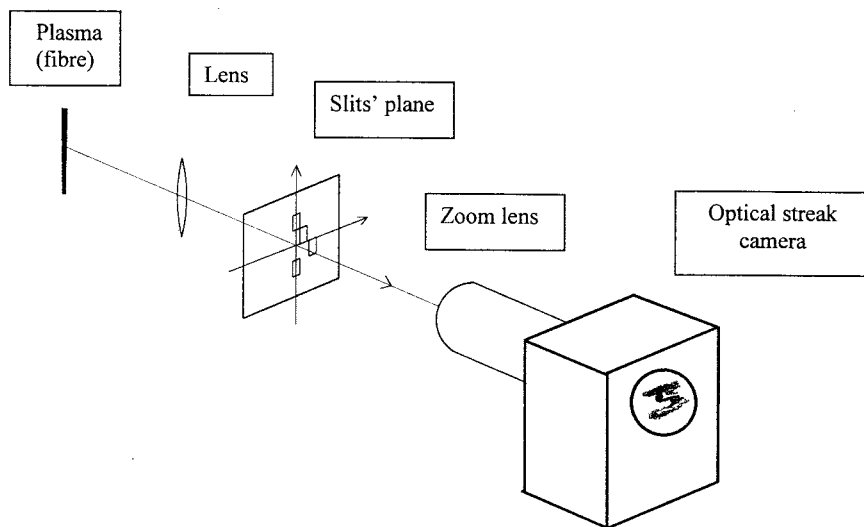


Fig. 2. Schematic of the multislit optical streak arrangement.

paths through the plasma. After crossing the plasma the beams were separated again. These two beams were used to take two Schlieren photographs 3 ns apart with a sensitivity of 4×10^{-4} radians. One of the two perpendicularly polarized beams was separated from the other and used to take an interferogram of the plasma using a self-referencing interferometer [20]. All the images from the laser diagnostic were recorded on charge coupled device (CCD) cameras.

Axial optical streak photography was performed to investigate the temporal evolution of the axial structure of the plasma. In some shots a set of four parallel-aligned slits (Fig. 2), with

their longitudinal axis parallel to the axis of the fiber (z -axis) was used. Each slit was 5 mm long and 0.55 mm wide, which gave a time resolution of 4 ns for the streak camera sweep used. A two-lens system was used to focus the image of the fiber onto the photo cathode of the streak camera. The slits were placed in the image plane of the first lens. The first and the fourth slit were aligned with the central part of the fiber, the second was displaced by 2.4 mm from the central axis, and the third was displaced by 1 mm.

For the detection of X-rays a gated 4 frame (2 ns gate, 9 ns separation) X-ray camera, used in conjunction with a set

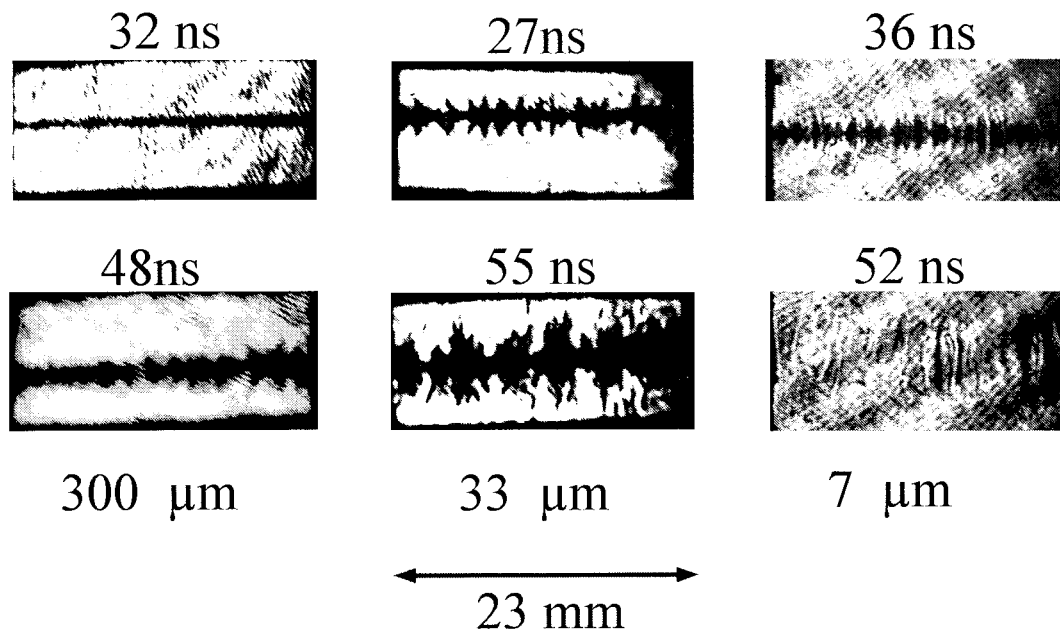


Fig. 3. Schlieren photographs of carbon fibers of 7, 33, and 300 μm diameters, taken without prepulse.

of filtered pinholes was employed. Pinholes of 100 μm and 300 μm were used, which gave a spatial resolution of 0.3 and 1 mm, respectively. The pinholes were filtered with either 10 μm Be, 1.5 μm aluminized mylar, or 1.5 μm polycarbonate.

III. EXPERIMENTAL RESULTS AND DISCUSSION

A. Discharges Without Current Prepulse

The Schlieren photographs were used to measure the average size of the plasma. Considering the sensitivity of the Schlieren setup (4×10^{-4} radian), the minimum density gradient detected with this system was $2 \times 10^{17} \text{ cm}^{-4}$. A set of Schlieren photographs taken in different shots from carbon fibers of 7, 33, and 300 μm (without using a current prepulse) is presented in Fig. 3. From these photographs the mean value of the diameter of the coronal plasma was measured and its average expansion velocity was calculated. For the 300 μm fibers, the velocity is about 1 $\text{cm}/\mu\text{s}$ and the fiber has a more homogeneous expansion in terms of the instabilities. For 7 and 33 μm fibers the velocity was about 5 $\text{cm}/\mu\text{s}$ but the former has a shorter life time, as can be seen in Fig. 3. At 50 ns, with a 7 μm fiber, the plasma diameter is about 5 mm and after that time the plasma density becomes below the detection limit even for the interferometer. The sensitivity of the interferometer is about $n l = 10^{17} \text{ cm}^{-2}$, where n is the electron density and l the path length of the laser beam through the plasma. For 33 μm fibers, the pinch column remains continuous up to about 200 ns, when density islands surrounded by low density regions are evident as is shown in Fig. 4. The appearance of these density islands can be associated with the presence of high energy electron beams [21].

Comparison of the data obtained from the different diagnostics in the same discharge is shown in Fig. 5. From the two frames of Schlieren pictures with 3 ns time separation it is possible to obtain information about the axial motion of

plasma. The photographs were taken about 30 ns after the current start. This is the time when the streak photograph shows axial motion of the emitting region [22] with a velocity of about $2 \times 10^7 \text{ cm/s}$. This velocity would give a displacement of 0.6 mm between the two frames, which is larger than the wavelength of the perturbations at this moment, but it is not observed in the Schlieren images. The typical spatial scale of the inhomogeneities in the optical streak images is two to four times larger than the wavelength of perturbations in the coronal plasma (from the Schlieren data). The X-ray gated images show a similar spatial scale to the optical streak. This is because X-ray and optical emission originate from the central dense part of the pinch, while the Schlieren images show the corona plasma with much less density.

Multiple-slit optical streak photographs taken with 33 and 300 μm fibers are shown in Fig. 6. From these photographs it can also be concluded that most of the visible emission comes from the central part of the fiber. It was found that the size of the emitting region is much smaller than that estimated using the Schlieren photographs. For example, at 35 ns the emitting region of the plasma is about 1.5 mm in diameter while the coronal plasma, as measured using the Schlieren photographs, is around 4 mm.

B. Effect of Current Prepulse

A set of shots was carried out to study the influence of the current prepulse on the coronal plasma of the Z-pinch initiated from a 33 μm carbon fiber. Fig. 7 shows a waveform of the current through the pinch, where time $t = 0$ corresponds to the start of the main current. The prepulse current reached 35 kA in about 200 ns. The amplitude of the main current with and without prepulse was essentially the same. The most significant difference between the two regimes (with and without prepulse) was the delay time between the voltage pulse and the current start (fiber breakdown). For shots without

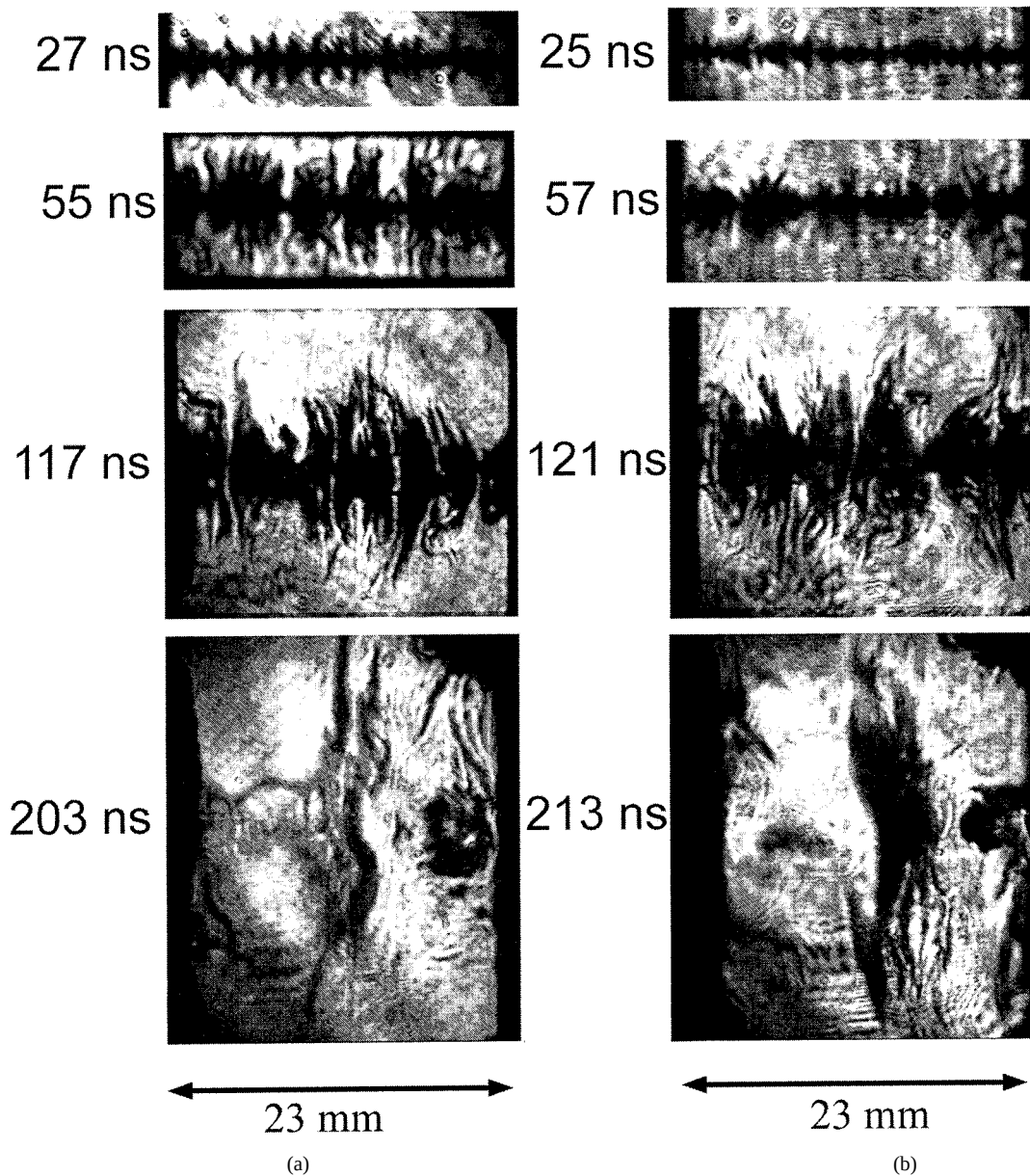


Fig. 4. Schlieren photographs of carbon fiber $33\text{ }\mu\text{m}$ diameter taken at different times in the discharge for shots with and without prepulse. The anode and the cathode are located at the right and left side, respectively.

prepulse the main current starts with a delay of 20–30 ns. The voltage at the cathode at this moment is about 200 kV. For shots with prepulse the main current starts practically without any delay. It is interesting to note that with the prepulse, when the voltage rise is slow ($\sim 1\text{ }\mu\text{s}$), breakdown occurs at the much lower voltage of about 20 kV.

Fig. 4 shows a set of representative Schlieren pictures at different times. Instabilities with $m = 0$ appear within the first 10 ns of the main current pulse. The instability pattern for both regimes does not change significantly between two Schlieren frames separated by 3 ns. Differences between the two regimes are observed only during the beginning of the discharge (up to about 80 ns, when the current is less than 250 kA), when it is found that the size of the coronal plasma is smaller with the prepulse. Later, after about 120 ns, there are practically no differences in the coronal plasma behavior and

variations from shot to shot become more significant. From the Schlieren pictures it is possible to measure the size (diameter) of perturbations and their wavelength. These measurements are shown in Fig. 8, each data point representing one shot, the error bars being the standard deviation of the measured value along the length of the fiber. The diameter of the perturbations increases linearly in time while the diameter of the necking region increases at a much lower rate. This explains the increase in time of the size of the error bars. The radial velocity is $(3.6 \pm 0.4) \times 10^6\text{ cm/s}$ with prepulse. Without prepulse the radial velocity is slightly larger $[(5.5 \pm 1.5) \times 10^6\text{ cm/s}]$. The wavelength (λ_z) of the perturbations is slowly increasing with time from 0.05 cm to about 0.2 cm in 100 ns and is about the same for both regimes. Also shown in Fig. 8 is the calculated $ka = \pi D/\lambda_z$, where a is the radius of perturbations and D is the measured coronal plasma diameter. With the prepulse the

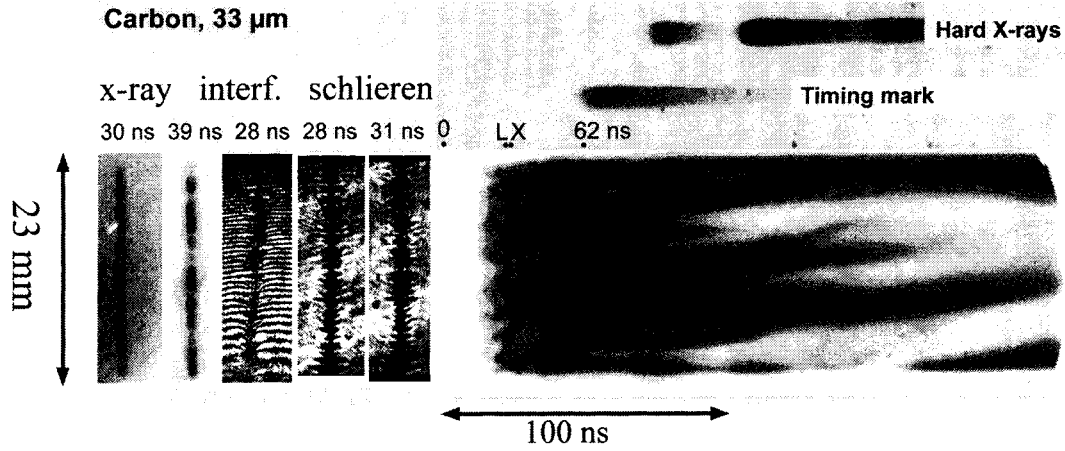


Fig. 5. X-ray, interferometry, Schlieren, and optical axial streak for a 33- μm fiber for a shot without prepulse. In the streak photograph, L is the time of the laser probing and X the time of the first frame of the X-ray camera. The cathode is at the bottom of the figure.

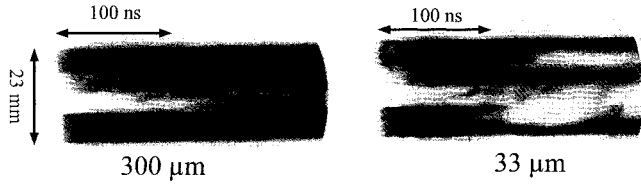


Fig. 6. Multislit optical streak for 33 and 300 μm fibers.

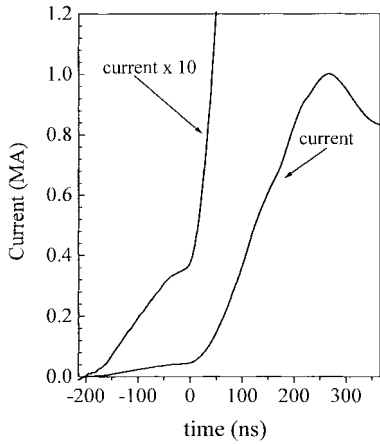


Fig. 7. Graph of the pinch current in a discharge taken with a prepulse. The current signal was also multiplied by ten in order to see the beginning of the rise current.

value of ka is about ten, while for shots without the prepulse ka is up to 20. The difference is mainly due to the faster expansion of the corona for the latter regime.

Another value of the expansion velocity of the coronal plasma is measured from Schlieren pictures obtained in the same shot with 3 ns time separation. This velocity agrees with the average value obtained from Fig. 8. However, the velocity varies along the length of the pinch, the maximum values being 2×10^7 cm/s. It is also possible to observe an inward motion in the neck regions with velocities $\sim 10^7$ cm/s.

The local Alfvén velocity (V_A) for the corona plasma can be calculated, if we assume that all current is flowing inside the corona plasma [20] and that the integral $\int (\nabla n_e) dl \approx n_e$.

Using the sensitivity of the Schlieren system results in an estimation of $n_e = 3 \times 10^{18} \text{ cm}^{-3}$. This gives $V_A = 2 \times 10^7$ cm/s for totally stripped carbon, which is two to four times higher than the measured average velocity of the coronal plasma expansion. Thus there could be a quasi-pressure balance in the coronal plasma. During the prepulse, the analysis of interferograms and Schlieren photographs shows that the plasma was formed along the whole length of the fiber with a diameter of ~ 0.5 mm and with line density $N_e \sim 5 \times 10^{17} \text{ m}^{-1}$. This corresponds to 0.5% of the initial line density of the carbon fiber with 33 μm diameter (assuming $Z = 1$). This plasma is not homogeneous along the fiber (Fig. 9). One or more regions with well-developed instability are usually observed. Optical emission recorded in the streak pictures shows that “zippering” starts from these places during the main current. The axial velocity of these emitting fronts for this shot is 3×10^7 cm/s. The gated X-ray pictures obtained in the same shot (Fig. 9) show two bright spots that move in opposite axial directions with the same velocity as the front of light emission.

IV. CONCLUSION

The expansion velocity of the corona plasma and the evolution of instability wavelengths were measured in carbon fiber pinches. Optical and X-ray emission, which comes mainly from the core plasma, shows longer wavelengths and different axial velocities than observed in the corona by laser probing. The spatial position of the X-ray hot spots coincides with the necks of $m = 0$ instabilities observed in Schlieren photographs, but not all the necks produce hot spots. The corona plasma in 7 and 33 μm fibers shows the same behavior until about 50 ns. After that time the density of the plasma formed from 7 μm fiber is below the detection level ($\sim 10^{17} \text{ cm}^{-3}$) while in the case of a 33 μm fiber, the pinch lasts up to about 200 ns, by which time it has evolved into a series of density islands. For a very thick fiber (300 μm), expansion of the corona plasma is much slower. This may be due to a lower temperature of the plasma in this case. A current prepulse provided initial ionization of the fiber pinch. A plasma with line density of $5 \times 10^{17} \text{ m}^{-1}$ (<1% of the initial fiber

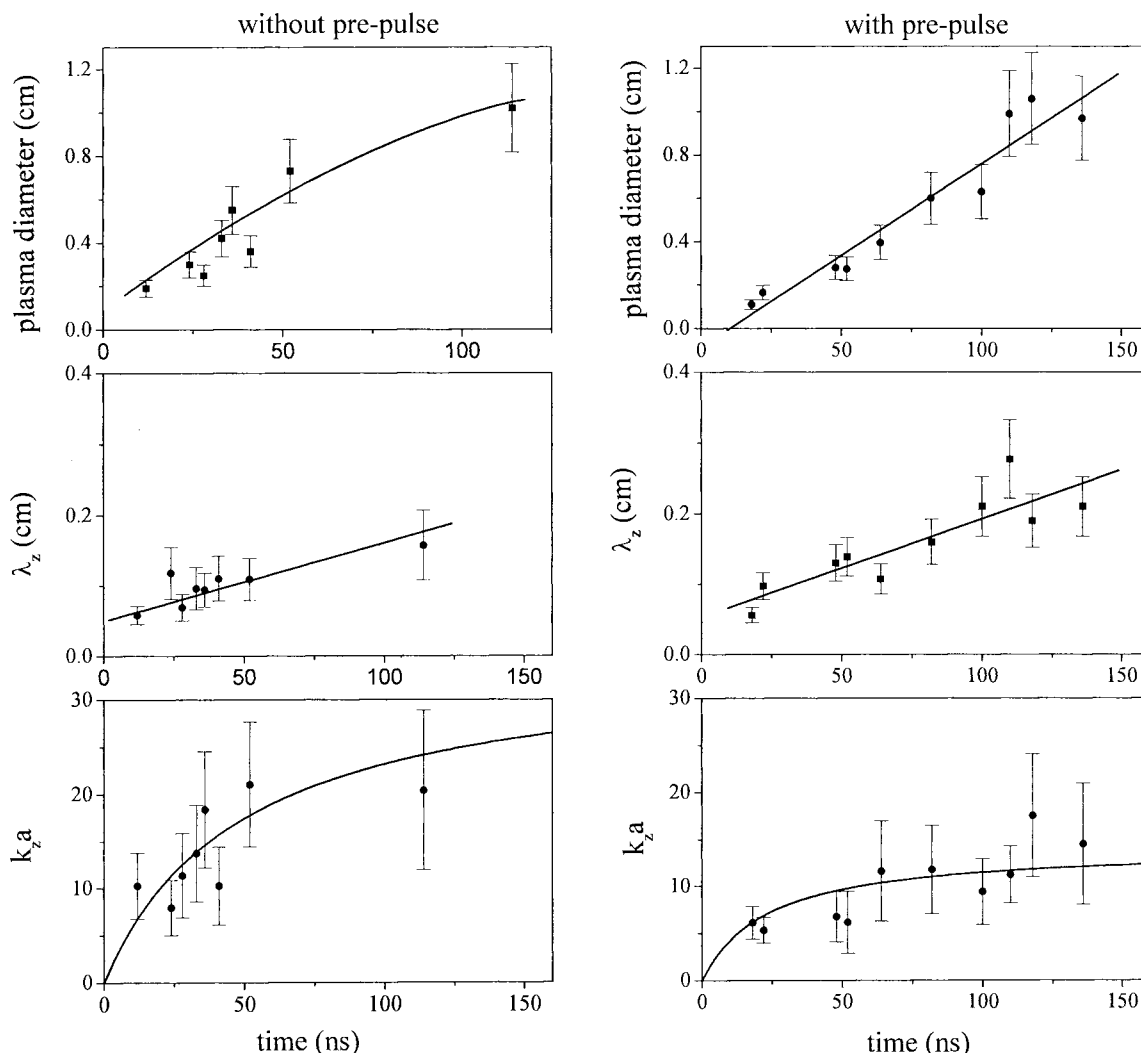


Fig. 8. Diameter of the corona plasma, instability wavelength, and ka product versus time for 33 μm fibers with and without prepulse.

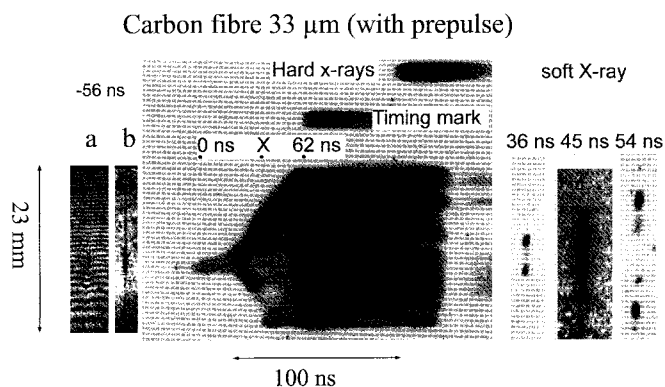


Fig. 9. (a) Interferometry and (b) Schlieren photographs taken 56 ns before the start of the main current. The X-ray gated images and axial optical streak photographs were taken during the main discharge.

line density) and diameter about 1 mm was produced before the start of the main current. It was found that in the regime with prepulse the expansion of the corona plasma was slower during the first 80 ns of the discharge but this did not affect the pinch stability later in time. It would be interesting to study other techniques of fiber pre-ionization, which would allow the

parameters of the plasma surrounding the fiber to be controlled before applying the pinch current.

REFERENCES

- [1] M. G. Haines, A. E. Dangor, A. Folkierski, P. Baldock, C. D. Challis, P. Choi, M. Coppins, C. Deeney, M. B. Favre-Dominguez, E. Figura, and J. D. Sethian, "Z-pinch experiments and theory," in *Plasma Physics and Controlled Nuclear Fusion Research*, vol. 2. Vienna, Austria: Int. Atomic Energy Agency, 1986, pp. 573–582.
- [2] M. G. Haines and M. Coppins, "Universal diagram for regimes of Z-pinch stability," *Phys. Rev. Lett.*, vol. 66, no. 11, pp. 1462–1465, 1991.
- [3] J. W. Shearer, "Contraction of Z-pinchs actuated by radiation losses," *Phys. Fluids*, vol. 19, no. 9, pp. 1426–1428, 1976.
- [4] V. V. Vikhrev, "Contraction of Z-pinch as a result of losses to radiation," *JETP Lett.*, vol. 27, no. 2, pp. 95–98, 1978.
- [5] M. G. Haines, "An analytical model of radiative collapse of a Z-pinch," *Plasma Phys. Control Fusion*, vol. 31, no. 5, pp. 759–778, 1989.
- [6] A. E. Robson, "Evolution of a Z-pinch with constant dI/dt ," *Nucl. Fusion*, vol. 28, no. 12, pp. 2171–2178, 1988.
- [7] R. S. Pease, "Equilibrium characteristics of a pinched gas discharge cooled by Bremsstrahlung radiation," in *Proc. Phys. Soc. London B*, vol. 70, pp. 11–23, 1957.
- [8] S. I. Braginskii, "The behavior of a completely ionized plasma in a strong magnetic field," *Sov. Phys.—JETP*, vol. 6, pp. 494–501, 1958.
- [9] I. R. Lindemuth, "Two-dimensional fiber ablation in the solid-deuterium Z-pinch," *Phys. Rev. Lett.*, vol. 65, no. 2, pp. 179–182, 1990.

- [10] J. P. Chittenden, R. Aliaga-Rossel, S. V. Lebedev, I. H. Mitchell, M. Tatarakis, A. R. Bell, and M. G. Haines, "Two-dimensional magneto-hydrodynamic modeling of carbon fiber Z-pinch experiments," *Phys. Plasmas*, vol. 4, no. 12, pp. 4309–4317, 1997.
- [11] D. Kalantar and D. Hammer, "Observation of a stable dense core within an unstable coronal plasma in wire-initiated dense Z-pinch experiments," *Phys. Rev. Lett.*, vol. 71, no. 23, pp. 3806–3809, 1993.
- [12] E. S. Figura, G. H. McCall, and A. E. Dangor, "Characteristic of a Z-pinch produced from a glass optical fiber," *Phys. Fluids B*, vol. 3, no. 3, pp. 2835–2843, 1991.
- [13] F. N. Beg, A. E. Dangor, P. Lee, M. Tatarakis, S. L. Niffikeer, and M. G. Haines, "Optical and X-ray observations of carbon and aluminum fiber Z-pinch plasmas," *Plasma Phys. Control Fusion*, vol. 39, no. 1, pp. 1–25, 1997.
- [14] R. Riley, D. Scudder, J. Shlachter, and R. Lovberg, "Instability heating of a solid fiber Z-pinch," *Phys. Plasma*, vol. 3, no. 4, pp. 1314–1323, 1996.
- [15] J. P. Chittenden, private communication, Aug. 1997.
- [16] D. W. Scudder, J. S. Shlachter, P. R. Forman, R. Riley, and R. H. Lovberg, "Fiber Z-pinch instabilities," in *Proc. 3rd Int. Conf. Dense Z-Pinches*, AIP Conf. Proc., London, U.K., 1993, no. 299, pp. 503–508.
- [17] A. Lorenz, F. N. Beg, J. Ruiz, J. F. Worley, and A. E. Dangor, "Carbon fiber Z-pinch with current prepulse," in *Proc. 4th Int. Conf. Dense Z-pinches*, AIP Conf. Proc., Vancouver, Canada, 1997, no. 409, pp. 315–318.
- [18] I. H. Mitchell, J. M. Bayley, J. P. Chittenden, J. F. Worley, A. E. Dangor, M. G. Haines, and P. Choi, "A high impedance mega-ampere generator for fiber Z-pinch experiments," *Rev. Sci. Instrum.*, vol. 57, no. 4, pp. 1533–1541, 1996.
- [19] R. Aliaga-Rossel, J. Bayley, A. Mamin, and Y. Nizienko, "SBS pulse compression applied to a commercial Q-switch Nd-YAG laser," in *Proc. 4th Int. Conf. Dense Z-pinches*, AIP Conf. Proc., Vancouver, Canada, 1997, no. 409, pp. 467–470.
- [20] M. Tatarakis, R. Aliaga-Rossel, A. E. Dangor, and M. G. Haines, "Optical probing of fiber Z-pinch plasmas," *Phys. Plasmas*, vol. 5, no. 3, pp. 682–691, 1998.
- [21] A. Robledo, I. H. Mitchell, R. Aliaga-Rossel, J. Chittenden, A. E. Dangor, and M. G. Haines, "Time-resolved energy measurements of electron beams in fiber Z-pinch discharges," *Plasma Phys.*, vol. 4, no. 2, pp. 490–492, 1997.
- [22] J. P. Chittenden, I. H. Mitchell, R. Aliaga-Rossel, J. M. Bayley, F. N. Beg, A. Lorenz, M. G. Haines, and G. Decker, "The dynamics of bifurcating bright-spots in fiber Z-pinch plasmas," *Phys. Plasmas*, vol. 4, no. 8, pp. 2967–2971, 1997.



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